# LINEAR AND ROBUST CONTROL SYSTEMS

## 11 - State Space to Minimal State Space by Decomposition Transformations

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## TOPICS

**1** SISO Transformations

**2** SISO Controllability Transform

**3** Decomposition Theorem

**4** MIMO Controllability Transform

**References**

Kailath chapter 6

## SS Transforamtions (SISO)

- We know $A, b, c^T \to \Lambda, \beta, \gamma^T$ via

$x_M = Q^T x =$ modal transform; $Q^T AP = \Lambda$

What of other transforms?

- For <u>minimal realizations</u> transforms easily found

$\mathcal{C}_T = T\mathcal{C} \Rightarrow T = \mathcal{C}_T \mathcal{C}^{-1}$

- Kailath[+] p.129 gives

(here $T$ takes $A, b, c^T$ to $A_T, b_T, c_T^T$)

Controllability form $\mathcal{C}_{co} = I$

$\Rightarrow T_{co} = I\mathcal{C}^{-1} = \mathcal{C}^{-1}$

Observability form $\mathcal{C}_{ob} = H_{n,n}$

$\Rightarrow T_{ob} = H_{n,n}\mathcal{C}^{-1} = \mathcal{O}$

Controller form $\mathcal{C}_c = \mathcal{A}_-^T$

$\Rightarrow T_c = \mathcal{A}_-^T \mathcal{C}^{-1}$

Observer form $\mathcal{C}_o = b(A_0)\tilde{I}$

$\Rightarrow T_o = b(A_o)\tilde{I}\mathcal{C}^{-1} = \mathcal{A}_- \mathcal{O}$ (prove it!)

[+]Kailath's $T$=my$T^{-1}$

## Direct derivation of controllability transform from a minimal SS model

$\hat{x} = \mathcal{C}^{-1}x$

$\Rightarrow A, b, c^T \rightarrow \mathcal{C}^{-1}A\mathcal{C}, \mathcal{C}^{-1}b, c^T\mathcal{C}$

Now we calculate

$\mathcal{C}^{-1}A\mathcal{C} = \mathcal{C}^{-1}(Ab, A^2b, \cdots, A^nb)$

But$\mathcal{C}^{-1}\mathcal{C} = I$

$$\begin{aligned} \Rightarrow \mathcal{C}^{-1}(b, Ab, \cdots, A^{n-1}b) &= (e_1, \cdots, e_n) \\ \Rightarrow \mathcal{C}^{-1}A^{i-1}b &= e_i \\ \Rightarrow \mathcal{C}^{-1}A\mathcal{C} = (e_2, e_3, \cdots, e_{n-1}, \mathcal{C}^{-1}A^nb) &= A_{co} \end{aligned}$$

## Direct derivation of controllability transform Continued

But Cayley-Hamilton gives

$$\begin{aligned}
A^n &= -a_1 A^{n-1} - a_2 A^{n-2} - \cdots - a_n I \\
\Rightarrow \mathcal{C}^{-1} A^n b &= -a_1 e_n - a_2 e_{n-1} - \cdots - a_n e_1 \\
&= \begin{pmatrix} -a_n \\ -a_{n-1} \\ \vdots \\ \vdots \\ -a_1 \end{pmatrix}
\end{aligned}$$

Hence $A_{co}$ has expected form.

Also

$$\begin{aligned}
\mathcal{C}^{-1} b &= e_1 \\
c^T \mathcal{C} &= (c^T b, c^T A b, \cdots, c^T A^{n-1} b) \\
&= (h_1, h_2, \cdots, h_n)
\end{aligned}$$

as required.

## Decomposition Theorem

If rank $\mathcal{C}(A, B) = r < n$

There is a transformation $T$ such that

$A, B, C^T \rightarrow \overline{A}, \overline{B}, \overline{C}^T$

$$\overline{A} = \begin{pmatrix} \overline{A}_c & * \\ 0 & \overline{A}_{\overline{c}} \end{pmatrix}, \overline{B} = \begin{pmatrix} \overline{B}_c \\ 0 \end{pmatrix}$$

$$, \overline{C}^T = \left[ \begin{array}{cc} \overline{C}_c^T, & \overline{C}_{\overline{c}}^T \end{array} \right]$$

i) $\overline{A}_c, \overline{B}_c$ is controllable

ii) $\overline{P}(s) = \overline{C}_c^T (sI - \overline{A}_c)^{-1} \overline{B}_c = P(s)$

Similar result holds for observability.

Putting the two together gives a 4 point decomposition of the state space.

## DECOMPOSITION - PROOF

Since $\mathcal{C}(A, B)$ has rank $r$ we can find $r$ linearly independent columns. Collect them into a matrix $Z_{n\times r}$. Let $Q_{n\times(n-r)}$ consist of $n-r$ linearly independent column vectors orthogonal to the columns of $Z$, i.e. $Q^T Z = 0$

Then set

$$T = \begin{array}{cc} \left[\begin{array}{c} (Z^T Z)^{-1} Z^T \\ (Q^T Q)^{-1} Q^T \end{array}\right] & \begin{array}{c} (r) \\ (n-r) \end{array} \end{array}$$

By the projection lemma

$T^{-1} = [Z, Q]$

## DECOMPOSITION PROOF Continued

By column interchange bring $Z$ to occupy the first $r$ columns. Since the remaining columns are linearly dependent on the columns of $Z$ , they can be written as $ZW$. So we have

$$\mathcal{C}(A, B) = \begin{bmatrix} Z & ZW \end{bmatrix} J$$

where $J$ is a permutation matrix.

The transformed controllability matrix is thus

$$\begin{aligned}
\mathcal{C}_T &= T\mathcal{C} \\
&= \begin{bmatrix} (Z^T Z)^{-1} Z^T \\ (Q^T Q)^{-1} Q^T \end{bmatrix} \begin{bmatrix} Z, & ZW \end{bmatrix} J \\
&= \begin{pmatrix} I & W \\ 0 & 0 \end{pmatrix} \begin{pmatrix} J_a & J_b \\ J_c & J_d \end{pmatrix}, say \\
&= \begin{array}{cc} \begin{bmatrix} J_a + W J_c & J_b + W J_d \\ 0 & 0 \end{bmatrix} & \begin{array}{c} (r) \\ (n-r) \end{array} \end{array}
\end{aligned}$$

Note bottom rows are zeroed.

## DECOMPOSITION PROOF Continued II

Continuing, we can also write

$$
\begin{aligned}
\mathcal{C}_T &= T\mathcal{C} \\
&= \begin{bmatrix} (Z^T Z)^{-1} Z^T \\ (Q^T Q)^{-1} Q^T \end{bmatrix} \mathcal{C} \\
&= \begin{array}{c c} \begin{bmatrix} (Z^T Z)^{-1} Z^T \mathcal{C} \\ (Q^T Q)^{-1} Q^T \mathcal{C} \end{bmatrix} & \begin{matrix} (r) \\ (n-r) \end{matrix} \end{array}
\end{aligned}
$$

## DECOMPOSITION PROOF Continued III

Matching this to the expression above gives

$$
\begin{aligned}
0 &= Q^T\mathcal{C} \\
\Rightarrow 0 &= Q^T[B, AB, \cdots, A^{n-1}B] \\
\Rightarrow 0 &= Q^T A^{i-1}B = 0, i = 1, \cdots, n \\
\Rightarrow 0 &= Q^T A^n B, \text{ by Cayley-Hamilton} \\
\Rightarrow 0 &= Q^T[AB, \cdots, A^n B] \\
\Rightarrow 0 &= Q^T A[B, AB, \cdots, A^{n-1}B] \\
\Rightarrow 0 &= Q^T A\mathcal{C} \\
\Rightarrow 0 &= Q^T A[Z, ZW]J \\
\Rightarrow 0 &= [Q^T AZ, Q^T AZW] \\
\Rightarrow 0 &= Q^T AZ
\end{aligned}
$$

## DECOMPOSITION PROOF Continued IV

- We now find

$$
\begin{aligned}
A_T &= TAT^{-1} \\
&= \begin{bmatrix} (Z^TZ)^{-1}Z^T \\ (Q^TQ)^{-1}Q^T \end{bmatrix} A[Z,Q] \\
&= \begin{bmatrix} (Z^TZ)^{-1}Z^TAZ & * \\ 0 & (Q^TQ)^{-1}Q^TAQ \end{bmatrix} \\
&= \begin{bmatrix} \overline{A}_c & * \\ 0 & \overline{A}_{\overline{c}} \end{bmatrix}, \text{ as required}
\end{aligned}
$$

$$
\begin{aligned}
\bullet B_T &= TB \\
&= \begin{bmatrix} (Z^TZ)^{-1}Z^T \\ (Q^TQ)^{-1}Q^T \end{bmatrix} B \\
&= \begin{bmatrix} (Z^TZ)^{-1}Z^TB \\ 0 \end{bmatrix} = \begin{bmatrix} \overline{B}_c \\ 0 \end{bmatrix}
\end{aligned}
$$

as claimed.

# DECOMPOSITION PROOF Continued V

- $$\begin{aligned} C_T^T &= C^T T^{-1} \\ &= C^T [Z, Q] \\ &= [C^T Z, C^T Q] \\ &= [\overline{C}_c^T, \overline{C}_{\overline{c}}^T] \end{aligned}$$

- $$\begin{aligned} & C^T (sI - A)^{-1} B \\ = \;& (\overline{C}_c^T, \overline{C}_{\overline{c}}^T) \begin{pmatrix} sI - \overline{A}_c & * \\ 0 & sI - \overline{A}_{\overline{c}} \end{pmatrix}^{-1} \begin{pmatrix} \overline{B}_c \\ 0 \end{pmatrix} \\ = \;& (\overline{C}_c^T, \overline{C}_{\overline{c}}^T) \begin{pmatrix} \left(sI - \overline{A}_c\right)^{-1} & * \\ 0 & \left(sI - \overline{A}_{\overline{c}}\right)^{-1} \end{pmatrix} \begin{pmatrix} \overline{B}_c \\ 0 \end{pmatrix} \\ = \;& \left[\overline{C}_c^T (sI - \overline{A}_c)^{-1}, \overline{C}_{\overline{c}}^T (sI - \overline{A}_{\overline{c}})^{-1} + *\right] \begin{bmatrix} \overline{B}_c \\ 0 \end{bmatrix} \\ = \;& \overline{C}_c^T (sI - \overline{A}_c)^{-1} \overline{B}_c \end{aligned}$$

as required.

## SISO - Controllability form from non minimal SS form

$\mathcal{C} = [b, Ab, \cdots, A^{n-1}b]$

With $\mathcal{C}$ of rank $r < n$, the first $r$ columns must be linearly independent (why?)

So we can put

$Z = [b, Ab, \cdots, A^{r-1}b]$

Then

$$
\begin{aligned}
T &= \begin{bmatrix} (Z^T Z)^{-1} Z^T \\ (Q^T Q)^{-1} Q^T \end{bmatrix} = \begin{bmatrix} T_Z \\ T_Q \end{bmatrix} \\
T_Z Z &= I_r \\
\Rightarrow I_r &= T_Z (b, Ab, \cdots, A^{r-1}b) \\
\Rightarrow e_s &= T_Z A^{s-1} b, s = 1, \cdots, r
\end{aligned}
$$

## SISO transforms Continued

We can now calculate $\overline{A}_c, \overline{B}_c$

$$\begin{aligned}
\overline{A}_c &= (Z^T Z)^{-1} Z^T A Z = T_Z A Z \\
&= T_Z (Ab, A^2 b, \cdots, A^{r-1} b, A^r b) \\
&= [e_2, \cdots, e_r, T_Z A^r b]
\end{aligned}$$

But $\mathcal{C}(A, B)$ has rank $r$ so there are coefficients $\alpha_1, \cdots, \alpha_r$ with

$$\begin{aligned}
A^r b &= \sum_1^r \alpha_{r-s+1} A^{s-1} b \\
\Rightarrow T_Z A^r b &= -\sum_1^r \alpha_{r-s+1} T_Z A^{s-1} b \\
&= -\sum_1^r \alpha_{r-s+1} e_s
\end{aligned}$$

## SISO transforms Continued II

$$
= \begin{bmatrix} -\alpha_r \\ -\alpha_{r-1} \\ : \\ : \\ -\alpha_1 \end{bmatrix}
$$

$$
\Rightarrow \overline{A}_c = \begin{bmatrix} 0 & 0 & \cdots & -\alpha_r \\ 1 & 0 & \cdots & -\alpha_{r-1} \\ 0 & 1 & \cdots & : \\ : & : & \cdots & : \\ : & : & \cdots & : \\ 0 & 0 & 1 & -\alpha_1 \end{bmatrix}
$$

which is in controllability form. Also

$$
\overline{B}_c = (Z^T Z)^{-1} Z^T b = T_Z b = e_1
$$

as required.

# MIMO Controllability Transform Crate Table

Need to search columns of $\mathcal{C}(A, B)$ to find linearly independent columns. Results clearly shown in 'crate' table.

| $b_1$ | $b_2$ | $b_3$ | $b_4$ | |
|---|---|---|---|---|
| $X$ | $X$ | $X$ | 0 | $I$ |
| $X$ | 0 | $X$ | | $A$ |
| $X$ | | 0 | | $A^2$ |
| 0 | | | | $A^3$ |

Fig.11A.1 Crate Table [$m = 4$]

$k_1 = 3, k_2 = 1, k_3 = 2$

$$\begin{aligned} \mathcal{C}(A, B) &= [B, AB, A^2B, \cdots] \\ &= [b_1, b_2, b_3, b_4, Ab_1, Ab_2, Ab_3, Ab_4, A^2b_1 \cdots] \end{aligned}$$

# MIMO Controllability Transform Crate Table Continued

Search across the crate, row by row.

Put an $X$ if the corresponding vector is linearly independent. Put a 0 if linearly dependent.

e.g. Fig. shows $b_4$ linearly dependent on $b_1, b_2, b_3$

Note then that $A^r b_4$ will be linearly dependent or $A^r b_1, A^r b_2, A^r b_3$ so need not look at cells below a 0

Second row shows $Ab_2$ linearly dependent on $b_1, b_2, b_3, Ab_1$

But $Ab_3$ is linearly independent of $b_1, b_2, b_3, Ab_1$.

Highest powers $A^{k_i} b_i$ in the linearly independent set determine indices $k_i, i = 1, \cdots, m$

If we change the order of the $b_i$ in $B$

We still get some indices $k_i$

## MIMO Controllability Form

Can now construct minimal SS form.

In the example of Fig.11A.1 put

$$Z = [b_1, Ab_1, A^2b_1, b_2, b_3, Ab_3], \text{ note order}$$

Minimal SS dimension is

$$n_{\min} = r = \sum_1^m k_i = 3 + 1 + 2 + 0 = 6$$

Now as before

$$\begin{aligned} T_Z &= (Z^T Z)^{-1} Z^T \Rightarrow T_Z Z = I_{r\times r} \\ &\Rightarrow T_Z[b_1, Ab_1, A^2b_1, b_2, b_3, Ab_3] = [e_1, \cdots, e_6] \end{aligned}$$

Thus

$$\begin{aligned} \overline{A}_c &= (Z^T Z)^{-1} Z^T AZ = T_Z AZ \\ &= T_Z[Ab_1, A^2b_1, A^3b_1, Ab_2, Ab_3, A^2b_3] \\ &= [e_2, e_3, \xi_1 | \xi_2 | e_6, \xi_3] \end{aligned}$$

So we have 3 controllability subblocks.

## MIMO Controllability Form Continued

By linear dependence there are coefficient vectors with

$$
\begin{aligned}
A^3 b_1 &= -Z\alpha_1 \Rightarrow \xi_1 = T_Z A^3 b_1 = -T_Z Z \alpha_1 = -\alpha_1 \\
A b_2 &= -Z\alpha_2 \Rightarrow \xi_2 = T_Z A b_2 = -\alpha_2 \\
A^2 b_3 &= -Z\alpha_3 \Rightarrow \xi_3 = T_Z A^2 b_3 = -\alpha_3
\end{aligned}
$$

Then

$$
\overline{A}_c = \begin{bmatrix}
0 & 0 & -\alpha_{1,1} & -\alpha_{2,1} & 0 & -\alpha_{3,1} \\
1 & 0 & -\alpha_{1,2} & -\alpha_{2,2} & 0 & -\alpha_{3,2} \\
0 & 1 & -\alpha_{1,3} & -\alpha_{2,3} & 0 & -\alpha_{3,3} \\
0 & 0 & -\alpha_{1,4} & -\alpha_{2,4} & 0 & -\alpha_{3,4} \\
0 & 0 & -\alpha_{1,5} & -\alpha_{2,5} & 0 & -\alpha_{3,5} \\
0 & 0 & -\alpha_{1,6} & -\alpha_{2,6} & 1 & -\alpha_{3,6}
\end{bmatrix}
$$

## MIMO Controllability Form Continued II

$$\begin{aligned} \overline{B}_c &= T_Z B = T_Z(b_1, b_2, b_3, b_4) \\ &= (e_1, e_4, e_5, \zeta) \end{aligned}$$

Linear dependence

$$\begin{aligned} \Rightarrow b_4 &= -Z\beta_1 \Rightarrow \zeta = T_Z(-Z\beta_1) = -\beta_1 \\ \Rightarrow \overline{B}_c &= \begin{bmatrix} 1 & 0 & 0 & -\beta_{1,1} \\ 0 & 0 & 0 & -\beta_{1,2} \\ 0 & 0 & 0 & -\beta_{1,3} \\ 0 & 1 & 0 & -\beta_{1,4} \\ 0 & 0 & 1 & -\beta_{1,5} \\ 0 & 0 & 0 & -\beta_{1,6} \end{bmatrix} \end{aligned}$$

NB. Can find e.g. $\alpha_1$ as $\alpha_1 = -T_Z A^3 b_1$ i.e. by constructing $T_Z$ and applying it to $A^3 b_1$ etc.

## Example 11.1

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$C^T = [\ -1 \quad 0 \quad -1\ ], D = [\ 0, \quad 1\ ]$$

We find [clearly $n_{\min} \leq 3$]

$$\begin{aligned} \mathcal{C}(A,B) &= (B, AB, A^2B) \\ &= \begin{array}{c} {\scriptstyle b_1, b_2, Ab_1, Ab_2, A^2b_1, A^2b_2} \\ \begin{pmatrix} 1 & 0 & 1 & 2 & 14 & 20 \\ 0 & 1 & 2 & 3 & 20 & 29 \\ 0 & 0 & 3 & 4 & 26 & 28 \end{pmatrix} \end{array} \end{aligned}$$

## Example 11.1 Continued

Fill out crate by inspection

| | | |
|---|---|---|
| $b_1$ | $b_2$ | |
| $X$ | $X$ | $I$ |
| $X$ | $0$ | $A$ |
| $0$ | | $A^2$ |

$\Rightarrow k_1 = 2, k_2 = 1 \Rightarrow n_{\min} = k_1 + k_2 = 3$

$$\Rightarrow Z \quad = \quad b_1, Ab_1, b_2$$
$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 3 & 0 \end{pmatrix}$$

and $T_Z Z = T_Z(b_1, Ab_1, b_2) = [e_1, e_2, e_3]$

# Example11.1 Continued II

$$\begin{aligned}
\overline{A}_c &= T_Z A Z = T_Z(Ab_1, A^2 b_1, Ab_2) \\
&= (e_2, \xi_1, \xi_2) \\
&= (e_2, -\alpha_1, -\alpha_2) \\
\alpha_1 &= -T_Z A^2 b_1 \\
\alpha_2 &= -T_Z A b_2
\end{aligned}$$

We find

$$\begin{aligned}
T_Z &= (Z^T Z)^{-1} Z^T = Z^{-1} \\
&= \begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \end{bmatrix} \\
\Rightarrow & \quad (\alpha_1, \alpha_2) = - \begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 14 & 2 \\ 20 & 3 \\ 26 & 4 \end{bmatrix}
\end{aligned}$$

# Example11.1 Continued III

$$
\begin{aligned}
&= -\begin{bmatrix} 14-\frac{26}{3} & 2-\frac{4}{3} \\ \frac{26}{3} & \frac{4}{3} \\ 20-\frac{52}{3} & 3-\frac{8}{3} \end{bmatrix} \\
&= -\begin{bmatrix} 5\frac{1}{3} & \frac{2}{3} \\ 8\frac{2}{3} & 1\frac{1}{3} \\ 2\frac{2}{3} & \frac{1}{3} \end{bmatrix} \\
\Rightarrow \overline{A}_c &= \begin{bmatrix} 0 & +5\frac{1}{3} & \frac{2}{3} \\ 1 & +8\frac{2}{3} & 1\frac{1}{3} \\ 0 & +2\frac{2}{3} & \frac{1}{3} \end{bmatrix}
\end{aligned}
$$

Also

$$
\overline{B}_c = T_Z(b_1 b_2) = (e_1 e_3) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}
$$